

Falcon Tokenizer Evaluation Report

FLORES-200 · 63 LANGUAGES · 17 REGIONAL WINNERS

Generated: 2026-06-16 11:50 UTC · Source: `data/results.csv` · Corpus: FLORES-200 devtest (~1012 sentences/language) · Project: Flam AI — Falcon Language Support

Global Tokenizer Evaluation — Detailed Report

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Caveats. Fertility uses whitespace "words", which is imperfect for languages without whitespace word boundaries (Japanese, Thai, Burmese, Khmer). For those languages, use `avg_tokens_per_sent` as the primary metric instead of fertility. Vocabulary coverage is computed from characters present in the FLORES-200 corpus only.

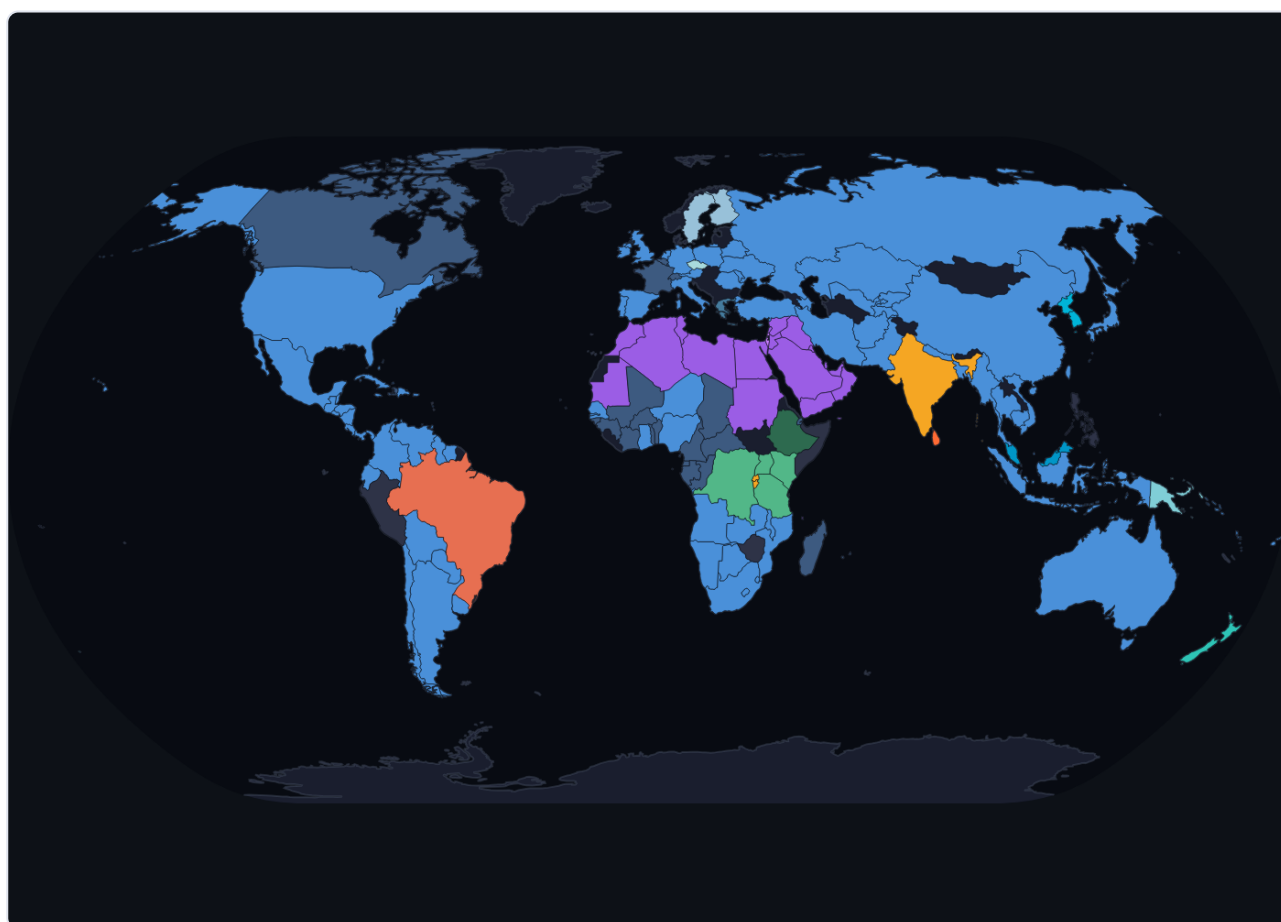
1. Best LLM per language — summary

For languages where the regional candidate tokenizer was not tested (gated repo, missing package, or no model found), Gemma-4 is the current default. Re-run after fixing those to get a full comparison.

#	Language	Region	Best tokenizer	Candidate tested	Key metrics
1	Hindi	Indic	Gemma-4	Airavata-7B	fertility 1.385→1.823, vcov 60.81%
2	Bengali	Indic	Gemma-4	BanglaLLama-3.1-8B	fertility 1.683→8.021, vcov 58.02%
3	Tamil	Indic	Tamil-Mistral-7B	Tamil-Mistral-7B	fertility 2.374→1.734, vcov 86.61%
4	Telugu	Indic	Gemma-4	Telugu-Llama2-7B	fertility 2.843→19.59, vcov 50.0%
5	Kannada	Indic	Ambari-7B	Ambari-7B	fertility 3.241→2.736, vcov 81.69%
6	Malayalam	Indic	Gemma-4	MalayaLLM-Gemma-9B	fertility 3.357→5.875, vcov 100.0%
7	Marathi	Indic	MahaMarathi-7B	MahaMarathi-7B	fertility 1.987→1.638, vcov 84.87%
8	Gujarati	Indic	Gujju-Llama-7B	Gujju-Llama-7B	fertility 2.415→2.033, vcov 82.98%
9	Punjabi	Indic	Gemma-4	Dhee-Qwen3-Punjabi-2B	fertility 2.82→7.758, vcov 67.33%
10	Odia	Indic	Gemma-4	Qwen-Odia-7B	fertility 4.852→13.654, vcov 56.44%
11	Assamese	Indic	Gemma-4	Goldfish-ASM-125M	fertility 2.819→1.42, vcov 93.17%
12	Urdu	Indic	Gemma-4	Qalb-1.0-8B	fertility 1.489→3.02, vcov 91.11%
13	Nepali	Indic	Gemma-4	NEPALI-LLM-9B	fertility 2.216→2.48, vcov 100.0%
14	Sinhala	Indic	Gemma-4	llama3-sinhala-8B	fertility 2.997→11.324, vcov 52.38%
15	Maithili	Indic	Gemma-4	No regional candidate tested	—
16	Arabic	Middle East	Jais-2-8B	Jais-2-8B	fertility 2.031→1.462, vcov 86.82%
17	Persian	Middle East	Gemma-4	Maral-7B	fertility 1.679→5.138, vcov 53.49%
18	Turkish	Middle East	Gemma-4	Trendyol-8B	fertility 2.109→2.547, vcov 100.0%

#	Language	Region	Best tokenizer	Candidate tested	Key metrics
19	Hebrew	Middle East	DictaLM-2.0-7B	DictaLM-2.0-7B	fertility 2.706→2.642, vcov 83.33%
20	Kurdish	Middle East	Gemma-4	Mistral-Nemo-Kurdish	fertility 2.358→2.486, vcov 96.19%
21	Azerbaijani	Middle East	Gemma-4	mGPT-Azerbaijani-1.3B	fertility 2.873→3.553, vcov 97.09%
22	Uzbek	Middle East	Gemma-4	Mistral-7B-Uz	fertility 2.839→3.515, vcov 100.0%
23	Kazakh	Middle East	Gemma-4	KazLLM-8B	fertility 3.184→4.831, vcov 91.3%
24	Chinese	East Asia	Gemma-4	ChatGLM3-6B	fertility 14.132→13.184, vcov 10.85%
25	Japanese	East Asia	Gemma-4	LLM-jp-3-13B	fertility 30.432→25.627, vcov 71.3%
26	Korean	East Asia	Polyglot-Ko-12B	Polyglot-Ko-12B	fertility 2.415→2.199, vcov 93.26%
27	Vietnamese	SEA	Gemma-4	Arcee-VyLinh-3B	fertility 1.21→1.29, vcov 100.0%
28	Thai	SEA	Gemma-4	Typhoon2-7B	fertility 10.084→16.315, vcov 98.77%
29	Indonesian	SEA	Gemma-4	Nusantara-7B	fertility 1.575→2.146, vcov 100.0%
30	Malay	SEA	MaLLaM-5B	MaLLaM-5B	fertility 1.632→1.417, vcov 84.76%
31	Tagalog	SEA	Gemma-4	No regional candidate tested	—
32	Burmese	SEA	Gemma-4	Burmese-GPT-1B	fertility 6.136→10.719, vcov 91.5%
33	Khmer	SEA	Gemma-4	PrahokBART-62M	fertility 12.747→14.511, vcov 75.14%
34	Swahili	Africa	Swahili-Gemma-7B	Swahili-Gemma-7B	fertility 2.087→2.047, vcov 100.0%
35	Amharic	Africa	Walia-LLM-7B	Walia-LLM-7B	fertility 3.032→1.623, vcov 88.0%
36	Hausa	Africa	Gemma-4	HausaLlama-8B	fertility 1.864→2.13, vcov 93.46%
37	Yoruba	Africa	Gemma-4	YorubaLlama-8B	fertility 2.588→2.934, vcov 96.09%
38	Igbo	Africa	Gemma-4	Kakugo-3B-Igbo	fertility 2.358→2.688, vcov 95.0%
39	Zulu	Africa	Gemma-4	Xhosa-ZuluLlama3-8B	fertility 3.384→3.754, vcov 100.0%
40	Xhosa	Africa	Gemma-4	Xhosa-ZuluLlama3-8B	fertility 3.302→3.567, vcov 100.0%
41	Somali	Africa	Gemma-4	No regional candidate tested	—
42	Wolof	Africa	Gemma-4	Wolof-Qwen-1.5B	fertility 1.921→2.103, vcov 100.0%
43	Shona	Africa	Gemma-4	No regional candidate tested	—
44	French	Europe	Lucie-7B	Lucie-7B	fertility 1.49→1.427, vcov 82.2%
45	German	Europe	Gemma-4	LeoLM-7B	fertility 1.655→2.18, vcov 81.48%
46	Spanish	Europe	Gemma-4	Salamandra-7B	fertility 1.347→1.354, vcov 85.58%
47	Portuguese	Europe	Gemma-4	Gervasio-8B	fertility 1.453→1.709, vcov 100.0%
48	Italian	Europe	Gemma-4	LLaMAntino-3-8B	fertility 1.535→1.831, vcov 100.0%
49	Dutch	Europe	Gemma-4	Fietje-2	fertility 1.63→2.325, vcov 94.34%
50	Polish	Europe	Gemma-4	Bielik-11B	fertility 2.096→2.924, vcov 81.13%
51	Russian	Europe	Gemma-4	Vikhr-Nemo-12B	fertility 1.884→2.13, vcov 98.67%
52	Ukrainian	Europe	Gemma-4	MamayLM-12B	fertility 2.273→2.273, vcov 100.0%
53	Romanian	Europe	Gemma-4	LLMic-3B	fertility 1.8→2.192, vcov 77.68%
54	Swedish	Europe	Viking-7B	Viking-7B	fertility 1.841→1.474, vcov 100.0%
55	Czech	Europe	CSMPT-7B	CSMPT-7B	fertility 2.157→1.407, vcov 96.52%

#	Language	Region	Best tokenizer	Candidate tested	Key metrics
56	Greek	Europe	Meltemi-7B	Meltemi-7B	fertility 2.472→1.397, vcov 95.42%
57	Lat.Am. Spanish	Americas	Gemma-4	LatamGPT-70B	fertility 1.347→1.604, vcov 100.0%
58	Brazilian Portuguese	Americas	Tucano-2b4	Tucano-2b4	fertility 1.453→1.252, vcov 86.84%
59	Quechua	Americas	Gemma-4	No regional candidate tested	—
60	Nahuatl	Americas	—	Not in FLORES-200	—
61	Haitian Creole	Americas	Gemma-4	No regional candidate tested	—
62	Māori	Oceania	Goldfish-mri-39M	Goldfish-mri-39M	fertility 1.825→1.22, vcov 92.47%
63	Samoa	Oceania	Gemma-4	No regional candidate tested	—
64	Hawaiian	Oceania	—	Not in FLORES-200	—
65	Tok Pisin	Oceania	Goldfish-tpi-125M	Goldfish-tpi-125M	fertility 1.644→1.311, vcov 83.02%



Best LLM per country — 163 countries mapped

2. Metric glossary

All metrics computed over the same FLORES-200 devtest segments per language.

1.1 Quick reference: which direction is "better"?

Metric	Better	Target
Fertility	Lower	1.0–2.5 for script-based languages
Compression ratio	Higher	> 3.0 chars/token
Byte fallback rate (%)	Lower	0% ideal; any > 0 = unhandled script
UNK rate (%)	Lower	0%; any UNK = vocab miss
Vocabulary coverage (%)	Higher	> 80% of unique chars as single tokens
Roundtrip fidelity (%)	Higher	100% = lossless encode→decode
Avg tokens / segment	Lower	same story as fertility (inference cost)

1.2 Fertility

What it is: Tokens per whitespace-separated 'word': `total_tokens / total_words`.

Direction: Lower is better. High fertility = same document uses more tokens (higher API cost, less room in context).

1.3 Compression ratio

What it is: Characters per token: `total_chars / total_tokens`.

Direction: Higher is better. Low compression often means byte-level or very fine splitting.

1.4 Byte fallback rate

What it is: Percentage of output tokens that represent raw byte fallback, not normal subwords.

Direction: Lower is better. 0% means no detected byte-fallback tokens.

1.5 UNK rate

What it is: Percentage of token ids equal to the tokenizer's `unk_token_id`.

Direction: Lower is better; 0% is the target.

1.6 Vocabulary coverage

What it is: Among all unique characters in that language's test text, the fraction that encode to exactly **one** token when passed alone to the tokenizer.

Direction: Higher is better. High coverage means many script characters are first-class tokens.

1.7 Roundtrip fidelity

What it is: Per segment: after `encode` then `decode`, does the text match the original?

Direction: Higher is better; 100% means no lossy tokenization on this test set.

1.8 Avg tokens per segment

What it is: `total_tokens / total_sentences` .

Direction: Lower is better for cost/latency. Use this as the primary efficiency metric for non-whitespace languages (Japanese, Thai, Burmese, Khmer).

3. Aggregate summary (all languages)

Unweighted averages treat each language equally. **Character-weighted** averages weight by `total_chars` so languages with more text influence the score more.

Tokenizer	Languages tested	Avg fertility	Avg compression	Avg byte fallback %	Avg vocab coverage %	Avg roundtrip %
mT5	63	3.176	2.857	0.126	54.262	87.356
Gemma-4	63	3.217	3.027	0.298	99.672	100.0
Xhosa-ZuluLlama3-8B	2	3.66	2.506	0.2	100.0	100.0
BLOOM	63	4.548	2.961	2.698	93.277	100.0

4. Regional candidate vs Gemma-4 — verdict per language

Primary signals: fertility and vocab coverage. Secondary: byte fallback rate and roundtrip fidelity.

Language	Region	Candidate	Fertility G4	Fertility cand	Vcov G4 %	Vcov cand %	BFR cand %	RT cand %	Verdict
Bengali	Indic	BanglaLLama-3.1-8B	1.683	8.021	100.0	58.02	16.93	100.0	Gemma wins
Tamil	Indic	Tamil-Mistral-7B	2.374	1.734	100.0	86.61	0.0	100.0	Candidate wins
Telugu	Indic	Telugu-Llama2-7B	2.843	19.59	99.35	50.0	90.6	100.0	Gemma wins
Malayalam	Indic	MalayaLLM-Gemma-9B	3.357	5.875	100.0	100.0	0.0	100.0	Mixed
Gujarati	Indic	Gujju-Llama-7B	2.415	2.033	99.29	82.98	0.0	100.0	Candidate wins
Punjabi	Indic	Dhee-Qwen3-Punjabi-2B	2.82	7.758	99.33	67.33	19.43	43.38	Gemma wins
Urdu	Indic	Qalb-1.0-8B	1.489	3.02	100.0	91.11	17.51	100.0	Gemma wins
Nepali	Indic	NEPALI-LLM-9B	2.216	2.48	99.35	100.0	0.0	100.0	Mixed
Sinhala	Indic	llama3-sinhala-8B	2.997	11.324	98.64	52.38	28.31	100.0	Gemma wins
Persian	Middle East	Maral-7B	1.679	5.138	100.0	53.49	0.04	100.0	Mixed
Turkish	Middle East	Trendyol-8B	2.109	2.547	100.0	100.0	0.6	100.0	Mixed
Hebrew	Middle East	DictaLM-2.0-7B	2.706	2.642	100.0	83.33	0.0	100.0	Candidate wins
Kurdish	Middle East	Mistral-Nemo-Kurdish	2.358	2.486	100.0	96.19	0.51	100.0	Mixed
Uzbek	Middle East	Mistral-7B-Uz	2.839	3.515	97.85	100.0	0.0	100.0	Mixed
Japanese	East Asia	LLM-jp-3-13B	30.432	25.627	100.0	71.3	0.0	100.0	Mixed
Korean	East Asia	Polyglot-Ko-12B	2.415	2.199	99.35	93.26	0.13	100.0	Candidate wins
Vietnamese	SEA	Arcee-VyLinh-3B	1.21	1.29	100.0	100.0	0.89	100.0	Mixed
Thai	SEA	Typhoon2-7B	10.084	16.315	100.0	98.77	2.35	100.0	Gemma wins
Indonesian	SEA	Nusantara-7B	1.575	2.146	100.0	100.0	0.62	100.0	Mixed

Language	Region	Candidate	Fertility G4	Fertility cand	Vcov G4 %	Vcov cand %	BFR cand %	RT cand %	Verdict
Malay	SEA	MaLLaM-5B	1.632	1.417	100.0	84.76	0.04	100.0	Candidate wins
Burmese	SEA	Burmese-GPT-1B	6.136	10.719	100.0	91.5	0.6	100.0	Mixed
Swahili	Africa	Swahili-Gemma-7B	2.087	2.047	100.0	100.0	0.0	100.0	Candidate wins
Igbo	Africa	Kakugo-3B-Igbo	2.358	2.688	100.0	95.0	4.25	100.0	Gemma wins
Wolof	Africa	Wolof-Qwen-1.5B	1.921	2.103	99.19	100.0	0.53	100.0	Mixed
French	Europe	Lucie-7B	1.49	1.427	98.31	82.2	0.0	100.0	Candidate wins
German	Europe	LeoLM-7B	1.655	2.18	100.0	81.48	0.01	100.0	Mixed
Spanish	Europe	Salamandra-7B	1.347	1.354	100.0	85.58	0.03	100.0	Mixed
Portuguese	Europe	Gervasio-8B	1.453	1.709	100.0	100.0	0.74	100.0	Mixed
Italian	Europe	LLaMAntino-3-8B	1.535	1.831	100.0	100.0	0.58	100.0	Mixed
Dutch	Europe	Fietje-2	1.63	2.325	100.0	94.34	0.1	100.0	Mixed
Polish	Europe	Bielik-11B	2.096	2.924	100.0	81.13	3.58	100.0	Gemma wins
Russian	Europe	Vikhr-Nemo-12B	1.884	2.13	99.33	98.67	0.71	100.0	Mixed
Ukrainian	Europe	MamayLM-12B	2.273	2.273	100.0	100.0	0.0	100.0	Mixed
Romanian	Europe	LLMic-3B	1.8	2.192	100.0	77.68	16.17	4.15	Gemma wins
Swedish	Europe	Viking-7B	1.841	1.474	100.0	100.0	0.94	100.0	Candidate wins
Czech	Europe	CSMPT-7B	2.157	1.407	99.13	96.52	0.03	100.0	Candidate wins
Greek	Europe	Meltemi-7B	2.472	1.397	100.0	95.42	0.04	100.0	Candidate wins
Lat.Am. Spanish	Americas	LatamGPT-70B	1.347	1.604	100.0	100.0	0.65	100.0	Mixed
Brazilian Portuguese	Americas	Tucano-2b4	1.453	1.252	100.0	86.84	0.01	99.31	Candidate wins
Hindi	Indic	Airavata-7B	1.385	1.823	100.0	60.81	3.99	100.0	Gemma wins
Kannada	Indic	Ambari-7B	3.241	2.736	100.0	81.69	0.0	100.0	Candidate wins
Marathi	Indic	MahaMarathi-7B	1.987	1.638	100.0	84.87	0.0	100.0	

Language	Region	Candidate	Fertility G4	Fertility cand	Vcov G4 %	Vcov cand %	BFR cand %	RT cand %	Verdict
									Candidate wins
Odia	Indic	Qwen-Odia-7B	4.852	13.654	96.32	56.44	25.78	75.69	Gemma wins
Arabic	Middle East	Jais-2-8B	2.031	1.462	100.0	86.82	3.71	100.0	Candidate wins
Azerbaijani	Middle East	mGPT-Azerbaijani-1.3B	2.873	3.553	100.0	97.09	0.96	100.0	Mixed
Kazakh	Middle East	KazLLM-8B	3.184	4.831	100.0	91.3	8.27	100.0	Gemma wins
Chinese	East Asia	ChatGLM3-6B	14.132	13.184	99.86	10.85	0.02	100.0	Mixed
Khmer	SEA	PrahokBART-62M	12.747	14.511	98.38	75.14	0.0	0.0	Gemma wins
Amharic	Africa	Walia-LLM-7B	3.032	1.623	99.08	88.0	0.03	100.0	Candidate wins
Hausa	Africa	HausaLlama-8B	1.864	2.13	99.07	93.46	2.93	100.0	Gemma wins
Yoruba	Africa	YorubaLlama-8B	2.588	2.934	100.0	96.09	1.84	100.0	Gemma wins
Zulu	Africa	Xhosa-ZuluLlama3-8B	3.384	3.754	100.0	100.0	0.16	100.0	Mixed
Xhosa	Africa	Xhosa-ZuluLlama3-8B	3.302	3.567	99.03	100.0	0.24	100.0	Mixed
Assamese	Indic	Goldfish-ASM-125M	2.819	1.42	99.38	93.17	0.0	25.89	Mixed
Māori	Oceania	Goldfish-mri-39M	1.825	1.22	100.0	92.47	0.12	99.7	Candidate wins
Tok Pisin	Oceania	Goldfish-tpi-125M	1.644	1.311	100.0	83.02	0.0	98.42	Candidate wins

5. Comparison across metrics (pivot tables)

Rows = languages. Columns = tokenizers. Scan across a row to compare models on one language; scan down a column to see one model across languages.

Fertility (tokens / whitespace word)

Lower is better.

Indic

language	Gemma-4	BLOOM	mT5	Airavata-7B	BanglaLLama-3.1-8B	Tamil-Mistral-7B	Telugu-Llama2-7B	Ambari-7B	MalayaL
Hindi	1.385	1.37	2.086	1.823	—	—	—	—	—
Bengali	1.683	1.652	2.715	—	8.021	—	—	—	—
Tamil	2.374	2.099	2.541	—	—	1.734	—	—	—
Telugu	2.843	2.15	2.813	—	—	—	19.59	—	—
Kannada	3.241	2.217	3.007	—	—	—	—	2.736	—
Malayalam	3.357	2.54	3.059	—	—	—	—	—	5.875
Marathi	1.987	1.754	2.699	—	—	—	—	—	—
Gujarati	2.415	1.776	2.781	—	—	—	—	—	—
Punjabi	2.82	1.528	2.743	—	—	—	—	—	—
Odia	4.852	1.912	5.33	—	—	—	—	—	—
Assamese	2.819	1.999	3.372	—	—	—	—	—	—
Urdu	1.489	1.344	1.843	—	—	—	—	—	—
Nepali	2.216	1.723	2.638	—	—	—	—	—	—
Sinhala	2.997	10.689	2.666	—	—	—	—	—	—
Maithili	1.806	1.734	2.382	—	—	—	—	—	—

Middle East

language	Gemma-4	BLOOM	mT5	Jais-2-8B	Maral-7B	Trendyol-8B	DictaLM-2.0-7B	Mistral-Nemo-Kurdish	mGPT-Azerbaijani
Arabic	2.031	1.603	2.309	1.462	—	—	—	—	—
Persian	1.679	2.055	1.919	—	5.138	—	—	—	—
Turkish	2.109	3.051	2.146	—	—	2.547	—	—	—
Hebrew	2.706	4.568	2.363	—	—	—	2.642	—	—
Kurdish	2.358	2.446	2.108	—	—	—	—	2.486	—
Azerbaijani	2.873	3.296	2.383	—	—	—	—	—	3.553
Uzbek	2.839	2.974	2.556	—	—	—	—	—	—
Kazakh	3.184	5.089	2.336	—	—	—	—	—	—

East Asia

language	Gemma-4	BLOOM	mT5	ChatGLM3-6B	LLM-jp-3-13B	Polyglot-Ko-12B
Chinese	14.132	12.117	14.796	13.184	—	—
Japanese	30.432	44.828	28.237	—	25.627	—
Korean	2.415	4.887	2.781	—	—	2.199

SEA

language	Gemma-4	BLOOM	mT5	Arcee-VyLinh-3B	Typhoon2-7B	Nusantara-7B	MaLLaM-5B	Burmese-GPT-1B	PrahokBAR
Vietnamese	1.21	1.128	2.147	1.29	—	—	—	—	—
Thai	10.084	29.119	7.763	—	16.315	—	—	—	—
Indonesian	1.575	1.333	1.834	—	—	2.146	—	—	—
Malay	1.632	1.444	1.851	—	—	—	1.417	—	—
Tagalog	1.84	1.895	1.838	—	—	—	—	—	—
Burmese	6.136	25.746	4.947	—	—	—	—	10.719	—
Khmer	12.747	29.492	8.165	—	—	—	—	—	14.511

Africa

language	Gemma-4	BLOOM	mT5	Swahili-Gemma-7B	Walia-LLM-7B	HausaLIama-8B	YorubaLIama-8B	Kakugo-3B-Igbo	Xhosa
Swahili	2.087	1.602	1.972	2.047	—	—	—	—	—
Amharic	3.032	7.905	3.326	—	1.623	—	—	—	—
Hausa	1.864	1.924	1.82	—	—	2.13	—	—	—
Yoruba	2.588	1.759	2.755	—	—	—	2.934	—	—
Igbo	2.358	1.892	2.45	—	—	—	—	2.688	—
Zulu	3.384	3.043	2.99	—	—	—	—	—	3.754
Xhosa	3.302	2.933	2.913	—	—	—	—	—	3.567
Somali	2.246	2.402	2.159	—	—	—	—	—	—
Wolof	1.921	1.845	1.957	—	—	—	—	—	—
Shona	2.927	2.848	2.619	—	—	—	—	—	—

Europe

language	Gemma-4	BLOOM	mT5	Lucie-7B	LeoLM-7B	Salamandra-7B	Gervasio-8B	LLaMAntino-3-8B	Fietje-2	Bielik-11B
French	1.49	1.289	1.849	1.427	—	—	—	—	—	—
German	1.655	2.12	1.841	—	2.18	—	—	—	—	—
Spanish	1.347	1.269	1.696	—	—	1.354	—	—	—	—
Portuguese	1.453	1.312	1.858	—	—	—	1.709	—	—	—
Italian	1.535	1.828	1.864	—	—	—	—	1.831	—	—
Dutch	1.63	2.04	1.71	—	—	—	—	—	2.325	—
Polish	2.096	2.99	2.261	—	—	—	—	—	—	2.924
Russian	1.884	3.425	2.156	—	—	—	—	—	—	—
Ukrainian	2.273	3.899	2.325	—	—	—	—	—	—	—

language	Gemma-4	BLOOM	mT5	Lucie-7B	LeoLM-7B	Salamandra-7B	Gervasio-8B	LLaMAntino-3-8B	Fietje-2	Bielik-11B
Romanian	1.8	2.205	1.938	—	—	—	—	—	—	—
Swedish	1.841	2.222	1.849	—	—	—	—	—	—	—
Czech	2.157	2.885	2.242	—	—	—	—	—	—	—
Greek	2.472	4.341	2.323	—	—	—	—	—	—	—

Americas

language	Gemma-4	BLOOM	mT5	LatamGPT-70B	Tucano-2b4
Lat.Am. Spanish	1.347	1.269	1.696	1.604	—
Brazilian Portuguese	1.453	1.312	1.858	—	1.252
Quechua	3.147	3.158	3.017	—	—
Haitian Creole	1.796	1.85	1.772	—	—

Oceania

language	Gemma-4	BLOOM	mT5	Goldfish-mri-39M	Goldfish-tpi-125M
Māori	1.825	1.91	1.892	1.22	—
Samoan	1.821	1.823	2.029	—	—
Tok Pisin	1.644	1.668	1.766	—	1.311

Compression ratio (chars / token)

Higher is better.

Indic

language	Gemma-4	BLOOM	mT5	Airavata-7B	BanglaLLama-3.1-8B	Tamil-Mistral-7B	Telugu-LLama2-7B	Ambari-7B	MalayaL
Hindi	3.69	3.73	2.45	2.803	—	—	—	—	—
Bengali	3.93	4.004	2.436	—	0.825	—	—	—	—
Tamil	3.87	4.378	3.616	—	—	5.297	—	—	—
Telugu	2.751	3.637	2.78	—	—	—	0.399	—	—
Kannada	2.645	3.868	2.852	—	—	—	—	3.133	—
Malayalam	2.98	3.937	3.269	—	—	—	—	—	1.703
Marathi	3.517	3.982	2.588	—	—	—	—	—	—
Gujarati	2.506	3.407	2.176	—	—	—	—	—	—
Punjabi	1.827	3.372	1.878	—	—	—	—	—	—
Odia	1.416	3.595	1.29	—	—	—	—	—	—
Assamese	2.316	3.266	1.936	—	—	—	—	—	—
Urdu	3.156	3.496	2.549	—	—	—	—	—	—
Nepali	3.062	3.938	2.572	—	—	—	—	—	—
Sinhala	2.112	0.592	2.374	—	—	—	—	—	—
Maithili	2.912	3.032	2.208	—	—	—	—	—	—

Middle East

language	Gemma-4	BLOOM	mT5	Jais-2-8B	Maral-7B	Trendyol-8B	DictaLM-2.0-7B	Mistral-Nemo-Kurdish	mGPT-Azerbaijani
Arabic	2.917	3.695	2.565	4.052	—	—	—	—	—
Persian	3.135	2.56	2.742	—	1.024	—	—	—	—
Turkish	3.655	2.527	3.593	—	—	3.027	—	—	—
Hebrew	2.172	1.286	2.487	—	—	—	2.224	—	—
Kurdish	2.438	2.35	2.728	—	—	—	—	2.313	—
Azerbaijani	2.619	2.283	3.157	—	—	—	—	—	2.118
Uzbek	2.888	2.757	3.209	—	—	—	—	—	—
Kazakh	2.452	1.534	3.341	—	—	—	—	—	—

East Asia

language	Gemma-4	BLOOM	mT5	ChatGLM3-6B	LLM-jp-3-13B	Polyglot-Ko-12B
Chinese	1.479	1.725	1.413	1.585	—	—
Japanese	1.731	1.175	1.866	—	2.055	—
Korean	1.77	0.875	1.537	—	—	1.944

SEA

language	Gemma-4	BLOOM	mT5	Arcee-VyLinh-3B	Typhoon2-7B	Nusantara-7B	MaLLaM-5B	Burmese-GPT-1B	PrahokBAR
Vietnamese	3.727	3.998	2.101	3.496	—	—	—	—	—
Thai	2.906	1.006	3.775	—	1.796	—	—	—	—
Indonesian	4.562	5.39	3.918	—	—	3.347	—	—	—
Malay	4.437	5.014	3.914	—	—	—	5.109	—	—
Tagalog	3.374	3.275	3.377	—	—	—	—	—	—
Burmese	2.499	0.596	3.1	—	—	—	—	1.43	—
Khmer	2.067	0.893	3.227	—	—	—	—	—	1.816

Africa

language	Gemma-4	BLOOM	mT5	Swahili-Gemma-7B	Walia-LLM-7B	HausaLlama-8B	YorubaLlama-8B	Kakugo-3B-Igbo	Xhosa
Swahili	3.108	4.048	3.289	3.169	—	—	—	—	—
Amharic	1.658	0.636	1.512	—	3.098	—	—	—	—
Hausa	2.987	2.893	3.06	—	—	2.614	—	—	—
Yoruba	1.944	2.862	1.827	—	—	—	1.715	—	—
Igbo	2.29	2.854	2.204	—	—	—	—	2.009	—
Zulu	2.77	3.081	3.135	—	—	—	—	—	2.497
Xhosa	2.717	3.059	3.08	—	—	—	—	—	2.515
Somali	2.896	2.707	3.012	—	—	—	—	—	—
Wolof	2.643	2.751	2.594	—	—	—	—	—	—
Shona	2.907	2.987	3.249	—	—	—	—	—	—

Europe

language	Gemma-4	BLOOM	mT5	Lucie-7B	LeoLM-7B	Salamandra-7B	Gervasio-8B	LLaMAntino-3-8B	Fietje-2	Bielik-11B
French	4.129	4.774	3.328	4.311	—	—	—	—	—	—
German	4.27	3.333	3.839	—	3.242	—	—	—	—	—
Spanish	4.47	4.745	3.549	—	—	4.447	—	—	—	—
Portuguese	4.215	4.669	3.297	—	—	—	3.585	—	—	—
Italian	4.185	3.514	3.446	—	—	—	—	3.508	—	—
Dutch	3.935	3.144	3.75	—	—	—	—	—	2.759	—
Polish	3.415	2.394	3.166	—	—	—	—	—	—	2.447
Russian	3.838	2.111	3.353	—	—	—	—	—	—	—
Ukrainian	3.083	1.798	3.015	—	—	—	—	—	—	—
Romanian	3.479	2.84	3.23	—	—	—	—	—	—	—
Swedish	3.543	2.935	3.528	—	—	—	—	—	—	—
Czech	3.077	2.3	2.96	—	—	—	—	—	—	—
Greek	2.666	1.518	2.837	—	—	—	—	—	—	—

Americas

language	Gemma-4	BLOOM	mT5	LatamGPT-70B	Tucano-2b4
Lat.Am. Spanish	4.47	4.745	3.549	3.752	—
Brazilian Portuguese	4.215	4.669	3.297	—	4.893
Quechua	2.811	2.801	2.932	—	—
Haitian Creole	2.918	2.831	2.957	—	—

Oceania

language	Gemma-4	BLOOM	mT5	Goldfish-mri-39M	Goldfish-tpi-125M
Māori	2.647	2.529	2.554	3.959	—
Samoaan	2.635	2.632	2.366	—	—
Tok Pisin	3.247	3.201	3.024	—	4.071

Byte fallback rate (%)

Lower is better.

Indic

language	Gemma-4	BLOOM	mT5	Airavata-7B	BanglaLLama-3.1-8B	Tamil-Mistral-7B	Telugu-Llama2-7B	Ambari-7B	MalayaLL
Hindi	0.0	0.03	0.0	3.99	—	—	—	—	—
Bengali	0.0	0.27	0.0	—	16.93	—	—	—	—
Tamil	0.0	0.24	0.0	—	—	0.0	—	—	—
Telugu	1.0	0.62	0.0	—	—	—	90.6	—	—
Kannada	0.0	0.86	0.0	—	—	—	—	0.0	—

language	Gemma-4	BLOOM	mT5	Airavata-7B	BanglaLLama-3.1-8B	Tamil-Mistral-7B	Telugu-Llama2-7B	Ambari-7B	MalayaLL
Malayalam	0.0	0.17	0.0	—	—	—	—	—	0.0
Marathi	0.0	0.23	0.0	—	—	—	—	—	—
Gujarati	0.01	0.34	0.0	—	—	—	—	—	—
Punjabi	0.01	0.07	0.0	—	—	—	—	—	—
Odia	1.06	0.22	0.0	—	—	—	—	—	—
Assamese	0.01	0.26	0.0	—	—	—	—	—	—
Urdu	0.0	0.04	0.0	—	—	—	—	—	—
Nepali	0.0	0.04	0.0	—	—	—	—	—	—
Sinhala	0.02	30.3	0.0	—	—	—	—	—	—
Maithili	0.0	0.01	0.0	—	—	—	—	—	—

Middle East

language	Gemma-4	BLOOM	mT5	Jais-2-8B	Maral-7B	Trendyol-8B	DictaLM-2.0-7B	Mistral-Nemo-Kurdish	mGPT-Azerbaijani
Arabic	0.0	0.0	0.0	3.71	—	—	—	—	—
Persian	0.0	0.13	0.0	—	0.04	—	—	—	—
Turkish	0.96	0.12	0.19	—	—	0.6	—	—	—
Hebrew	0.0	7.34	0.0	—	—	—	0.0	—	—
Kurdish	0.0	0.09	0.0	—	—	—	—	0.51	—
Azerbaijani	0.91	0.47	0.12	—	—	—	—	—	0.96
Uzbek	0.01	0.0	0.0	—	—	—	—	—	—
Kazakh	0.0	2.79	0.0	—	—	—	—	—	—

East Asia

language	Gemma-4	BLOOM	mT5	ChatGLM3-6B	LLM-jp-3-13B	Polyglot-Ko-12B
Chinese	0.05	0.79	0.0	0.02	—	—
Japanese	0.0	2.32	0.0	—	0.0	—
Korean	0.07	22.1	0.0	—	—	0.13

SEA

language	Gemma-4	BLOOM	mT5	Arcee-VyLinh-3B	Typhoon2-7B	Nusantara-7B	MaLLaM-5B	Burmese-GPT-1B	PrahokBART
Vietnamese	0.18	0.02	0.15	0.89	—	—	—	—	—
Thai	0.0	11.86	0.0	—	2.35	—	—	—	—
Indonesian	0.0	0.0	0.0	—	—	0.62	—	—	—
Malay	0.0	0.0	0.0	—	—	—	0.04	—	—
Tagalog	0.0	0.02	0.0	—	—	—	—	—	—
Burmese	0.0	27.36	0.0	—	—	—	—	0.6	—
Khmer	0.01	26.35	0.0	—	—	—	—	—	0.0

Africa

language	Gemma-4	BLOOM	mT5	Swahili-Gemma-7B	Walia-LLM-7B	HausaLlama-8B	YorubaLlama-8B	Kakugo-3B-Igbo	Xhosa
Swahili	0.0	0.05	0.0	0.0	—	—	—	—	—
Amharic	0.02	26.31	0.0	—	0.03	—	—	—	—
Hausa	0.0	1.5	0.0	—	—	2.93	—	—	—
Yoruba	0.01	0.11	0.01	—	—	—	1.84	—	—
Igbo	0.0	0.15	0.0	—	—	—	—	4.25	—
Zulu	0.0	0.01	0.0	—	—	—	—	—	0.16
Xhosa	0.01	0.03	0.0	—	—	—	—	—	0.24
Somali	0.0	0.4	0.0	—	—	—	—	—	—
Wolof	0.02	0.01	0.0	—	—	—	—	—	—
Shona	0.0	0.02	0.0	—	—	—	—	—	—

Europe

language	Gemma-4	BLOOM	mT5	Lucie-7B	LeoLM-7B	Salamandra-7B	Gervasio-8B	LLaMAntino-3-8B	Fietje-2	Bielik-11B
French	2.04	0.0	0.0	0.0	—	—	—	—	—	—
German	0.01	0.05	0.0	—	0.01	—	—	—	—	—
Spanish	0.0	0.0	0.0	—	—	0.03	—	—	—	—
Portuguese	0.0	0.04	0.0	—	—	—	0.74	—	—	—
Italian	0.0	0.01	0.0	—	—	—	—	0.58	—	—
Dutch	0.01	0.0	0.0	—	—	—	—	—	0.1	—
Polish	1.94	0.03	1.52	—	—	—	—	—	—	3.58
Russian	0.01	0.24	0.0	—	—	—	—	—	—	—
Ukrainian	0.0	2.11	0.0	—	—	—	—	—	—	—
Romanian	2.4	1.54	1.39	—	—	—	—	—	—	—
Swedish	0.0	0.06	0.0	—	—	—	—	—	—	—
Czech	1.51	0.25	0.43	—	—	—	—	—	—	—
Greek	0.0	1.45	0.0	—	—	—	—	—	—	—

Americas

language	Gemma-4	BLOOM	mT5	LatamGPT-70B	Tucano-2b4
Lat.Am. Spanish	0.0	0.0	0.0	0.65	—
Brazilian Portuguese	0.0	0.04	0.0	—	0.01
Quechua	0.0	0.0	0.0	—	—
Haitian Creole	0.0	0.05	0.0	—	—

Oceania

language	Gemma-4	BLOOM	mT5	Goldfish-mri-39M	Goldfish-tpi-125M
Māori	5.77	0.02	3.89	0.12	—
Samoan	0.72	0.02	0.26	—	—

language	Gemma-4	BLOOM	mT5	Goldfish-mri-39M	Goldfish-tpi-125M
Tok Pisin	0.0	0.0	0.0	—	0.0

UNK rate (%)

Lower is better.

Indic

language	Gemma-4	BLOOM	mT5	Airavata-7B	BanglaLLama-3.1-8B	Tamil-Mistral-7B	Telugu-Llama2-7B	Ambari-7B	MalayaLL
Hindi	0.0	0.0	0.0	0.0	—	—	—	—	—
Bengali	0.0	0.0	0.0	—	0.0	—	—	—	—
Tamil	0.0	0.0	0.0	—	—	0.0	—	—	—
Telugu	0.0	0.0	0.0	—	—	—	0.0	—	—
Kannada	0.0	0.0	0.0	—	—	—	—	0.0	—
Malayalam	0.0	0.0	0.0	—	—	—	—	—	0.0
Marathi	0.0	0.0	0.0	—	—	—	—	—	—
Gujarati	0.0	0.0	0.0	—	—	—	—	—	—
Punjabi	0.0	0.0	0.0	—	—	—	—	—	—
Odia	0.0	0.0	0.0	—	—	—	—	—	—
Assamese	0.0	0.0	0.0	—	—	—	—	—	—
Urdu	0.0	0.0	0.0	—	—	—	—	—	—
Nepali	0.0	0.0	0.0	—	—	—	—	—	—
Sinhala	0.0	0.0	0.0	—	—	—	—	—	—
Maithili	0.0	0.0	0.0	—	—	—	—	—	—

Middle East

language	Gemma-4	BLOOM	mT5	Jais-2-8B	Maral-7B	Trendyol-8B	DictaLM-2.0-7B	Mistral-Nemo-Kurdish	mGPT-Azerbaijani
Arabic	0.0	0.0	0.0	0.0	—	—	—	—	—
Persian	0.0	0.0	0.0	—	0.0	—	—	—	—
Turkish	0.0	0.0	0.0	—	—	0.0	—	—	—
Hebrew	0.0	0.0	0.0	—	—	—	0.0	—	—
Kurdish	0.0	0.0	0.0	—	—	—	—	0.0	—
Azerbaijani	0.0	0.0	0.0	—	—	—	—	—	0.0
Uzbek	0.0	0.0	0.0	—	—	—	—	—	—
Kazakh	0.0	0.0	0.0	—	—	—	—	—	—

East Asia

language	Gemma-4	BLOOM	mT5	ChatGLM3-6B	LLM-jp-3-13B	Polyglot-Ko-12B
Chinese	0.0	0.0	0.02	0.0	—	—
Japanese	0.0	0.0	0.0	—	0.0	—

language	Gemma-4	BLOOM	mT5	ChatGLM3-6B	LLM-jp-3-13B	Polyglot-Ko-12B
Korean	0.0	0.0	0.02	—	—	0.0

SEA

language	Gemma-4	BLOOM	mT5	Arcee-VyLinh-3B	Typhoon2-7B	Nusantara-7B	MaLLaM-5B	Burmese-GPT-1B	PrahokBART
Vietnamese	0.0	0.0	0.0	0.0	—	—	—	—	—
Thai	0.0	0.0	0.0	—	0.0	—	—	—	—
Indonesian	0.0	0.0	0.0	—	—	0.0	—	—	—
Malay	0.0	0.0	0.0	—	—	—	0.0	—	—
Tagalog	0.0	0.0	0.0	—	—	—	—	—	—
Burmese	0.0	0.0	0.0	—	—	—	—	0.0	—
Khmer	0.0	0.0	0.0	—	—	—	—	—	8.97

Africa

language	Gemma-4	BLOOM	mT5	Swahili-Gemma-7B	Walia-LLM-7B	HausaLlama-8B	YorubaLlama-8B	Kakugo-3B-Igbo	Xhosa
Swahili	0.0	0.0	0.0	0.0	—	—	—	—	—
Amharic	0.0	0.0	0.0	—	0.0	—	—	—	—
Hausa	0.0	0.0	0.0	—	—	0.0	—	—	—
Yoruba	0.0	0.0	0.0	—	—	—	0.0	—	—
Igbo	0.0	0.0	0.0	—	—	—	—	0.0	—
Zulu	0.0	0.0	0.0	—	—	—	—	—	0.0
Xhosa	0.0	0.0	0.0	—	—	—	—	—	0.0
Somali	0.0	0.0	0.0	—	—	—	—	—	—
Wolof	0.0	0.0	0.0	—	—	—	—	—	—
Shona	0.0	0.0	0.0	—	—	—	—	—	—

Europe

language	Gemma-4	BLOOM	mT5	Lucie-7B	LeoLM-7B	Salamandra-7B	Gervasio-8B	LLaMAntino-3-8B	Fietje-2	Bielik-11B
French	0.0	0.0	0.0	0.0	—	—	—	—	—	—
German	0.0	0.0	0.0	—	0.0	—	—	—	—	—
Spanish	0.0	0.0	0.0	—	—	0.0	—	—	—	—
Portuguese	0.0	0.0	0.0	—	—	—	0.0	—	—	—
Italian	0.0	0.0	0.0	—	—	—	—	0.0	—	—
Dutch	0.0	0.0	0.0	—	—	—	—	—	0.0	—
Polish	0.0	0.0	0.0	—	—	—	—	—	—	0.0
Russian	0.0	0.0	0.0	—	—	—	—	—	—	—
Ukrainian	0.0	0.0	0.0	—	—	—	—	—	—	—
Romanian	0.0	0.0	0.0	—	—	—	—	—	—	—
Swedish	0.0	0.0	0.0	—	—	—	—	—	—	—

language	Gemma-4	BLOOM	mT5	Lucie-7B	LeoLM-7B	Salamandra-7B	Gervasio-8B	LLaMAntino-3-8B	Fietje-2	Bielik-11B
Czech	0.0	0.0	0.0	—	—	—	—	—	—	—
Greek	0.0	0.0	0.0	—	—	—	—	—	—	—

Americas

language	Gemma-4	BLOOM	mT5	LatamGPT-70B	Tucano-2b4
Lat.Am. Spanish	0.0	0.0	0.0	0.0	—
Brazilian Portuguese	0.0	0.0	0.0	—	0.0
Quechua	0.0	0.0	0.0	—	—
Haitian Creole	0.0	0.0	0.0	—	—

Oceania

language	Gemma-4	BLOOM	mT5	Goldfish-mri-39M	Goldfish-tpi-125M
Māori	0.0	0.0	0.0	0.0	—
Samoan	0.0	0.0	0.0	—	—
Tok Pisin	0.0	0.0	0.0	—	0.04

Vocabulary coverage (%)

Higher is better.

Indic

language	Gemma-4	BLOOM	mT5	Airavata-7B	BanglaLLama-3.1-8B	Tamil-Mistral-7B	Telugu-Llama2-7B	Ambari-7B	MalayaL
Hindi	100.0	98.65	54.73	60.81	—	—	—	—	—
Bengali	100.0	99.38	52.47	—	58.02	—	—	—	—
Tamil	100.0	100.0	59.06	—	—	86.61	—	—	—
Telugu	99.35	99.35	59.74	—	—	—	50.0	—	—
Kannada	100.0	100.0	55.63	—	—	—	—	81.69	—
Malayalam	100.0	98.57	62.86	—	—	—	—	—	100.0
Marathi	100.0	100.0	59.21	—	—	—	—	—	—
Gujarati	99.29	95.74	56.03	—	—	—	—	—	—
Punjabi	99.33	99.33	56.67	—	—	—	—	—	—
Odia	96.32	96.32	37.42	—	—	—	—	—	—
Assamese	99.38	98.76	55.28	—	—	—	—	—	—
Urdu	100.0	98.52	53.33	—	—	—	—	—	—
Nepali	99.35	100.0	57.79	—	—	—	—	—	—
Sinhala	98.64	52.38	51.02	—	—	—	—	—	—
Maithili	100.0	99.34	57.24	—	—	—	—	—	—

Middle East

language	Gemma-4	BLOOM	mT5	Jais-2-8B	Maral-7B	Trendyol-8B	DictaLM-2.0-7B	Mistral-Nemo-Kurdish	mGPT-Azerbaijani
Arabic	100.0	100.0	53.49	86.82	—	—	—	—	—
Persian	100.0	99.22	56.59	—	53.49	—	—	—	—
Turkish	100.0	98.15	55.56	—	—	100.0	—	—	—
Hebrew	100.0	96.3	62.04	—	—	—	83.33	—	—
Kurdish	100.0	99.05	58.1	—	—	—	—	96.19	—
Azerbaijani	100.0	97.09	63.11	—	—	—	—	—	97.09
Uzbek	97.85	98.92	64.52	—	—	—	—	—	—
Kazakh	100.0	86.96	67.08	—	—	—	—	—	—

East Asia

language	Gemma-4	BLOOM	mT5	ChatGLM3-6B	LLM-jp-3-13B	Polyglot-Ko-12B
Chinese	99.86	98.06	3.35	10.85	—	—
Japanese	100.0	87.24	4.2	—	71.3	—
Korean	99.35	19.36	33.58	—	—	93.26

SEA

language	Gemma-4	BLOOM	mT5	Arcee-VyLinh-3B	Typhoon2-7B	Nusantara-7B	MaLLaM-5B	Burmese-GPT-1B	PrahokBAR
Vietnamese	100.0	100.0	37.71	100.0	—	—	—	—	—
Thai	100.0	81.48	41.36	—	98.77	—	—	—	—
Indonesian	100.0	99.03	59.22	—	—	100.0	—	—	—
Malay	100.0	100.0	58.1	—	—	—	84.76	—	—
Tagalog	99.07	98.15	57.41	—	—	—	—	—	—
Burmese	100.0	53.59	50.98	—	—	—	—	91.5	—
Khmer	98.38	63.78	51.89	—	—	—	—	—	75.14

Africa

language	Gemma-4	BLOOM	mT5	Swahili-Gemma-7B	Walia-LLM-7B	HausaLlama-8B	YorubaLlama-8B	Kakugo-3B-Igbo	Xhosa
Swahili	100.0	100.0	61.96	100.0	—	—	—	—	—
Amharic	99.08	28.0	29.85	—	88.0	—	—	—	—
Hausa	99.07	93.46	56.07	—	—	93.46	—	—	—
Yoruba	100.0	99.22	53.91	—	—	—	96.09	—	—
Igbo	100.0	100.0	52.5	—	—	—	—	95.0	—
Zulu	100.0	99.09	59.09	—	—	—	—	—	100.0
Xhosa	99.03	99.03	59.22	—	—	—	—	—	100.0
Somali	100.0	100.0	60.4	—	—	—	—	—	—
Wolof	99.19	100.0	54.47	—	—	—	—	—	—
Shona	100.0	100.0	61.29	—	—	—	—	—	—

Europe

language	Gemma-4	BLOOM	mT5	Lucie-7B	LeoLM-7B	Salamandra-7B	Gervasio-8B	LLaMAntino-3-8B	Fietje-2	Bielik-11B
French	98.31	100.0	53.39	82.2	—	—	—	—	—	—
German	100.0	100.0	58.33	—	81.48	—	—	—	—	—
Spanish	100.0	100.0	59.62	—	—	85.58	—	—	—	—
Portuguese	100.0	100.0	53.51	—	—	—	100.0	—	—	—
Italian	100.0	100.0	56.6	—	—	—	—	100.0	—	—
Dutch	100.0	98.11	56.6	—	—	—	—	—	94.34	—
Polish	100.0	97.17	56.6	—	—	—	—	—	—	81.13
Russian	99.33	92.67	64.67	—	—	—	—	—	—	—
Ukrainian	100.0	88.44	63.27	—	—	—	—	—	—	—
Romanian	100.0	96.43	55.36	—	—	—	—	—	—	—
Swedish	100.0	99.05	60.0	—	—	—	—	—	—	—
Czech	99.13	95.65	54.78	—	—	—	—	—	—	—
Greek	100.0	86.27	62.75	—	—	—	—	—	—	—

Americas

language	Gemma-4	BLOOM	mT5	LatamGPT-70B	Tucano-2b4
Lat.Am. Spanish	100.0	100.0	59.62	100.0	—
Brazilian Portuguese	100.0	100.0	53.51	—	86.84
Quechua	100.0	100.0	60.0	—	—
Haitian Creole	100.0	99.13	59.13	—	—

Oceania

language	Gemma-4	BLOOM	mT5	Goldfish-mri-39M	Goldfish-tpi-125M
Māori	100.0	94.62	62.37	92.47	—
Samoan	100.0	97.37	54.39	—	—
Tok Pisin	100.0	100.0	58.49	—	83.02

Roundtrip fidelity (%)

Higher is better.

Indic

language	Gemma-4	BLOOM	mT5	Airavata-7B	BanglaLLama-3.1-8B	Tamil-Mistral-7B	Telugu-LLama2-7B	Ambari-7B	MalayaL
Hindi	100.0	100.0	90.32	100.0	—	—	—	—	—
Bengali	100.0	100.0	36.66	—	100.0	—	—	—	—
Tamil	100.0	100.0	92.49	—	—	100.0	—	—	—
Telugu	100.0	100.0	73.02	—	—	—	100.0	—	—
Kannada	100.0	100.0	54.55	—	—	—	—	100.0	—
Malayalam	100.0	100.0	71.15	—	—	—	—	—	100.0

language	Gemma-4	BLOOM	mT5	Airavata-7B	BanglaLLama-3.1-8B	Tamil-Mistral-7B	Telugu-Llama2-7B	Ambari-7B	MalayaL
Marathi	100.0	100.0	91.7	—	—	—	—	—	—
Gujarati	100.0	100.0	91.8	—	—	—	—	—	—
Punjabi	100.0	100.0	43.18	—	—	—	—	—	—
Odia	100.0	100.0	64.23	—	—	—	—	—	—
Assamese	100.0	100.0	26.38	—	—	—	—	—	—
Urdu	100.0	100.0	99.21	—	—	—	—	—	—
Nepali	100.0	100.0	94.17	—	—	—	—	—	—
Sinhala	100.0	100.0	30.43	—	—	—	—	—	—
Maithili	100.0	100.0	92.98	—	—	—	—	—	—

Middle East

language	Gemma-4	BLOOM	mT5	Jais-2-8B	Maral-7B	Trendyol-8B	DictaLM-2.0-7B	Mistral-Nemo-Kurdish	mGPT-Azerbaijani
Arabic	100.0	100.0	99.9	100.0	—	—	—	—	—
Persian	100.0	100.0	27.37	—	100.0	—	—	—	—
Turkish	100.0	100.0	99.31	—	—	100.0	—	—	—
Hebrew	100.0	100.0	98.32	—	—	—	100.0	—	—
Kurdish	100.0	100.0	99.41	—	—	—	—	100.0	—
Azerbaijani	100.0	100.0	99.7	—	—	—	—	—	100.0
Uzbek	100.0	100.0	99.41	—	—	—	—	—	—
Kazakh	100.0	100.0	98.62	—	—	—	—	—	—

East Asia

language	Gemma-4	BLOOM	mT5	ChatGLM3-6B	LLM-jp-3-13B	Polyglot-Ko-12B
Chinese	100.0	100.0	13.83	100.0	—	—
Japanese	100.0	100.0	89.23	—	100.0	—
Korean	100.0	100.0	98.52	—	—	100.0

SEA

language	Gemma-4	BLOOM	mT5	Arcee-VyLinh-3B	Typhoon2-7B	Nusantara-7B	MaLLaM-5B	Burmese-GPT-1B	PrahokBAR
Vietnamese	100.0	100.0	99.7	100.0	—	—	—	—	—
Thai	100.0	100.0	45.75	—	100.0	—	—	—	—
Indonesian	100.0	100.0	99.11	—	—	100.0	—	—	—
Malay	100.0	100.0	99.7	—	—	—	100.0	—	—
Tagalog	100.0	100.0	99.21	—	—	—	—	—	—
Burmese	100.0	100.0	96.25	—	—	—	—	100.0	—
Khmer	100.0	100.0	41.8	—	—	—	—	—	0.0

Africa

language	Gemma-4	BLOOM	mT5	Swahili-Gemma-7B	Walia-LLM-7B	HausaLlama-8B	YorubaLlama-8B	Kakugo-3B-Igbo	Xhos
Swahili	100.0	100.0	99.51	100.0	—	—	—	—	—
Amharic	100.0	100.0	99.31	—	100.0	—	—	—	—
Hausa	100.0	100.0	99.7	—	—	100.0	—	—	—
Yoruba	100.0	100.0	77.47	—	—	—	100.0	—	—
Igbo	100.0	100.0	94.47	—	—	—	—	100.0	—
Zulu	100.0	100.0	99.6	—	—	—	—	—	100.0
Xhosa	100.0	100.0	99.41	—	—	—	—	—	100.0
Somali	100.0	100.0	99.01	—	—	—	—	—	—
Wolof	100.0	100.0	99.51	—	—	—	—	—	—
Shona	100.0	100.0	99.8	—	—	—	—	—	—

Europe

language	Gemma-4	BLOOM	mT5	Lucie-7B	LeoLM-7B	Salamandra-7B	Gervasio-8B	LLaMAntino-3-8B	Fietje-2	Bielik-11B
French	100.0	100.0	96.84	100.0	—	—	—	—	—	—
German	100.0	100.0	98.91	—	100.0	—	—	—	—	—
Spanish	100.0	100.0	99.8	—	—	100.0	—	—	—	—
Portuguese	100.0	100.0	98.42	—	—	—	100.0	—	—	—
Italian	100.0	100.0	99.6	—	—	—	—	100.0	—	—
Dutch	100.0	100.0	99.7	—	—	—	—	—	100.0	—
Polish	100.0	100.0	99.6	—	—	—	—	—	—	100.0
Russian	100.0	100.0	95.95	—	—	—	—	—	—	—
Ukrainian	100.0	100.0	98.81	—	—	—	—	—	—	—
Romanian	100.0	100.0	99.7	—	—	—	—	—	—	—
Swedish	100.0	100.0	99.01	—	—	—	—	—	—	—
Czech	100.0	100.0	99.01	—	—	—	—	—	—	—
Greek	100.0	100.0	99.01	—	—	—	—	—	—	—

Americas

language	Gemma-4	BLOOM	mT5	LatamGPT-70B	Tucano-2b4
Lat.Am. Spanish	100.0	100.0	99.8	100.0	—
Brazilian Portuguese	100.0	100.0	98.42	—	99.31
Quechua	100.0	100.0	99.6	—	—
Haitian Creole	100.0	100.0	97.33	—	—

Oceania

language	Gemma-4	BLOOM	mT5	Goldfish-mri-39M	Goldfish-tpi-125M
Māori	100.0	100.0	99.8	99.7	—
Samoan	100.0	100.0	99.41	—	—

language	Gemma-4	BLOOM	mT5	Goldfish-mri-39M	Goldfish-tpi-125M
Tok Pisin	100.0	100.0	99.51	—	98.42

Avg tokens / segment

Lower is better.

Indic

language	Gemma-4	BLOOM	mT5	Airavata-7B	BanglaLLama-3.1-8B	Tamil-Mistral-7B	Telugu-Llama2-7B	Ambari-7B	Malaya
Hindi	35.1	34.72	52.86	46.2	—	—	—	—	—
Bengali	32.45	31.85	52.34	—	154.6	—	—	—	—
Tamil	39.36	34.79	42.12	—	—	28.75	—	—	—
Telugu	47.58	35.99	47.08	—	—	—	327.89	—	—
Kannada	51.56	35.27	47.83	—	—	—	—	43.53	—
Malayalam	49.52	37.48	45.14	—	—	—	—	—	86.67
Marathi	37.39	33.02	50.8	—	—	—	—	—	—
Gujarati	49.84	36.66	57.4	—	—	—	—	—	—
Punjabi	72.18	39.1	70.21	—	—	—	—	—	—
Odia	94.07	37.07	103.33	—	—	—	—	—	—
Assamese	53.73	38.11	64.26	—	—	—	—	—	—
Urdu	40.67	36.72	50.36	—	—	—	—	—	—
Nepali	40.96	31.84	48.76	—	—	—	—	—	—
Sinhala	61.33	218.73	54.56	—	—	—	—	—	—
Maithili	43.92	42.18	57.92	—	—	—	—	—	—

Middle East

language	Gemma-4	BLOOM	mT5	Jais-2-8B	Maral-7B	Trendyol-8B	DictaLM-2.0-7B	Mistral-Nemo-Kurdish	mGPT-Azerbaijani
Arabic	39.4	31.11	44.8	28.36	—	—	—	—	—
Persian	39.04	47.79	44.62	—	119.46	—	—	—	—
Turkish	36.7	53.08	37.34	—	—	44.31	—	—	—
Hebrew	46.44	78.4	40.55	—	—	—	45.35	—	—
Kurdish	52.84	54.81	47.22	—	—	—	—	55.7	—
Azerbaijani	54.24	62.23	44.99	—	—	—	—	—	67.07
Uzbek	51.09	53.51	45.99	—	—	—	—	—	—
Kazakh	54.58	87.25	40.05	—	—	—	—	—	—

East Asia

language	Gemma-4	BLOOM	mT5	ChatGLM3-6B	LLM-jp-3-13B	Polyglot-Ko-12B
Chinese	28.89	24.77	30.25	26.95	—	—
Japanese	32.51	47.88	30.16	—	27.37	—

language	Gemma-4	BLOOM	mT5	ChatGLM3-6B	LLM-jp-3-13B	Polyglot-Ko-12B
Korean	36.83	74.52	42.4	—	—	33.53

SEA

language	Gemma-4	BLOOM	mT5	Arcee-VyLinh-3B	Typhoon2-7B	Nusantara-7B	MaLLaM-5B	Burmese-GPT-1B	PrahokBAR
Vietnamese	36.75	34.26	65.18	39.17	—	—	—	—	—
Thai	42.88	123.81	33.01	—	69.37	—	—	—	—
Indonesian	30.83	26.09	35.89	—	—	42.02	—	—	—
Malay	32.56	28.81	36.92	—	—	—	28.28	—	—
Tagalog	48.66	50.13	48.62	—	—	—	—	—	—
Burmese	64.52	270.74	52.02	—	—	—	—	112.72	—
Khmer	73.25	169.46	46.92	—	—	—	—	—	83.38

Africa

language	Gemma-4	BLOOM	mT5	Swahili-Gemma-7B	Walia-LLM-7B	HausaLlama-8B	YorubaLlama-8B	Kakugo-3B-Igbo	Xhosa
Swahili	43.91	33.71	41.5	43.06	—	—	—	—	—
Amharic	51.93	135.37	56.95	—	27.79	—	—	—	—
Hausa	46.33	47.84	45.24	—	—	52.94	—	—	—
Yoruba	64.46	43.8	68.61	—	—	—	73.07	—	—
Igbo	58.05	46.58	60.33	—	—	—	—	66.19	—
Zulu	52.92	47.59	46.76	—	—	—	—	—	58.71
Xhosa	50.6	44.93	44.64	—	—	—	—	—	54.65
Somali	51.75	55.36	49.76	—	—	—	—	—	—
Wolof	47.54	45.68	48.44	—	—	—	—	—	—
Shona	50.18	48.83	44.9	—	—	—	—	—	—

Europe

language	Gemma-4	BLOOM	mT5	Lucie-7B	LeoLM-7B	Salamandra-7B	Gervasio-8B	LLaMAntino-3-8B	Fietje-2	Bielik-11B
French	37.73	32.63	46.8	36.13	—	—	—	—	—	—
German	35.59	45.59	39.59	—	46.88	—	—	—	—	—
Spanish	34.71	32.69	43.71	—	—	34.88	—	—	—	—
Portuguese	33.62	30.36	42.98	—	—	—	39.53	—	—	—
Italian	36.87	43.91	44.78	—	—	—	—	43.98	—	—
Dutch	37.05	46.37	38.87	—	—	—	—	—	52.83	—
Polish	40.37	57.58	43.53	—	—	—	—	—	—	56.32
Russian	36.58	66.52	41.88	—	—	—	—	—	—	—
Ukrainian	43.09	73.9	44.06	—	—	—	—	—	—	—
Romanian	42.2	51.69	45.45	—	—	—	—	—	—	—
Swedish	36.86	44.5	37.02	—	—	—	—	—	—	—

language	Gemma-4	BLOOM	mT5	Lucie-7B	LeoLM-7B	Salamandra-7B	Gervasio-8B	LLaMAntino-3-8B	Fietje-2	Bielik-11B
Czech	40.87	54.67	42.48	—	—	—	—	—	—	—
Greek	58.39	102.52	54.87	—	—	—	—	—	—	—

Americas

language	Gemma-4	BLOOM	mT5	LatamGPT-70B	Tucano-2b4
Lat.Am. Spanish	34.71	32.69	43.71	41.35	—
Brazilian Portuguese	33.62	30.36	42.98	—	28.96
Quechua	49.58	49.74	47.53	—	—
Haitian Creole	41.08	42.33	40.52	—	—

Oceania

language	Gemma-4	BLOOM	mT5	Goldfish-mri-39M	Goldfish-tpi-125M
Māori	54.7	57.26	56.7	36.57	—
Samoan	57.27	57.33	63.8	—	—
Tok Pisin	51.44	52.18	55.24	—	41.02

6. Complete per-language results

Every tokenizer × language combination from `data/results.csv`.

tokenizer	language	region	fertility	compression	byte_fallback%	unk%	vcov%	roundtrip%	avg_tok/sent	total
Gemma-4	Hindi	Indic	1.385	3.69	0.0	0.0	100.0	100.0	35.1	35.1
BLOOM	Hindi	Indic	1.37	3.73	0.03	0.0	98.65	100.0	34.72	35.1
Gemma-4	Bengali	Indic	1.683	3.93	0.0	0.0	100.0	100.0	32.45	32.45
BLOOM	Bengali	Indic	1.652	4.004	0.27	0.0	99.38	100.0	31.85	32.45
BanglaLLama-3.1-8B	Bengali	Indic	8.021	0.825	16.93	0.0	58.02	100.0	154.6	154.6
Gemma-4	Tamil	Indic	2.374	3.87	0.0	0.0	100.0	100.0	39.36	39.36
BLOOM	Tamil	Indic	2.099	4.378	0.24	0.0	100.0	100.0	34.79	35.1
Tamil-Mistral-7B	Tamil	Indic	1.734	5.297	0.0	0.0	86.61	100.0	28.75	29.0
Gemma-4	Telugu	Indic	2.843	2.751	1.0	0.0	99.35	100.0	47.58	48.0
BLOOM	Telugu	Indic	2.15	3.637	0.62	0.0	99.35	100.0	35.99	36.0
Telugu-Llama2-7B	Telugu	Indic	19.59	0.399	90.6	0.0	50.0	100.0	327.89	330.0
Gemma-4	Kannada	Indic	3.241	2.645	0.0	0.0	100.0	100.0	51.56	52.0
BLOOM	Kannada	Indic	2.217	3.868	0.86	0.0	100.0	100.0	35.27	35.0
Gemma-4	Malayalam	Indic	3.357	2.98	0.0	0.0	100.0	100.0	49.52	50.0
BLOOM	Malayalam	Indic	2.54	3.937	0.17	0.0	98.57	100.0	37.48	37.0
MalayaLLM-Gemma-9B	Malayalam	Indic	5.875	1.703	0.0	0.0	100.0	100.0	86.67	87.0
Gemma-4	Marathi	Indic	1.987	3.517	0.0	0.0	100.0	100.0	37.39	37.0
BLOOM	Marathi	Indic	1.754	3.982	0.23	0.0	100.0	100.0	33.02	33.0
Gemma-4	Gujarati	Indic	2.415	2.506	0.01	0.0	99.29	100.0	49.84	50.0
BLOOM	Gujarati	Indic	1.776	3.407	0.34	0.0	95.74	100.0	36.66	37.0
Gujju-Llama-7B	Gujarati	Indic	2.033	2.977	0.0	0.0	82.98	100.0	41.96	42.0
Gemma-4	Punjabi	Indic	2.82	1.827	0.01	0.0	99.33	100.0	72.18	73.0
BLOOM	Punjabi	Indic	1.528	3.372	0.07	0.0	99.33	100.0	39.1	39.0
Dhee-Qwen3-Punjabi-2B	Punjabi	Indic	7.758	0.664	19.43	0.0	67.33	43.38	198.6	200.0
Gemma-4	Odia	Indic	4.852	1.416	1.06	0.0	96.32	100.0	94.07	95.0
BLOOM	Odia	Indic	1.912	3.595	0.22	0.0	96.32	100.0	37.07	37.0
Gemma-4	Assamese	Indic	2.819	2.316	0.01	0.0	99.38	100.0	53.73	54.0
BLOOM	Assamese	Indic	1.999	3.266	0.26	0.0	98.76	100.0	38.11	38.0
Gemma-4	Urdu	Indic	1.489	3.156	0.0	0.0	100.0	100.0	40.67	41.0
BLOOM	Urdu	Indic	1.344	3.496	0.04	0.0	98.52	100.0	36.72	37.0
Qalb-1.0-8B	Urdu	Indic	3.02	1.556	17.51	0.0	91.11	100.0	82.52	83.0
Gemma-4	Nepali	Indic	2.216	3.062	0.0	0.0	99.35	100.0	40.96	41.0

tokenizer	language	region	fertility	compression	byte_fallback%	unk%	vcov%	roundtrip%	avg_tok/sent	tot
BLOOM	Nepali	Indic	1.723	3.938	0.04	0.0	100.0	100.0	31.84	32
NEPALI-LLM-9B	Nepali	Indic	2.48	2.736	0.0	0.0	100.0	100.0	45.83	46
Gemma-4	Sinhala	Indic	2.997	2.112	0.02	0.0	98.64	100.0	61.33	62
BLOOM	Sinhala	Indic	10.689	0.592	30.3	0.0	52.38	100.0	218.73	22
llama3-sinhala-8B	Sinhala	Indic	11.324	0.559	28.31	0.0	52.38	100.0	231.72	23
Gemma-4	Maithili	Indic	1.806	2.912	0.0	0.0	100.0	100.0	43.92	44
BLOOM	Maithili	Indic	1.734	3.032	0.01	0.0	99.34	100.0	42.18	42
Gemma-4	Arabic	Middle East	2.031	2.917	0.0	0.0	100.0	100.0	39.4	39
BLOOM	Arabic	Middle East	1.603	3.695	0.0	0.0	100.0	100.0	31.11	31
Gemma-4	Persian	Middle East	1.679	3.135	0.0	0.0	100.0	100.0	39.04	39
BLOOM	Persian	Middle East	2.055	2.56	0.13	0.0	99.22	100.0	47.79	48
Maral-7B	Persian	Middle East	5.138	1.024	0.04	0.0	53.49	100.0	119.46	12
Gemma-4	Turkish	Middle East	2.109	3.655	0.96	0.0	100.0	100.0	36.7	37
BLOOM	Turkish	Middle East	3.051	2.527	0.12	0.0	98.15	100.0	53.08	53
Trendyol-8B	Turkish	Middle East	2.547	3.027	0.6	0.0	100.0	100.0	44.31	44
Gemma-4	Hebrew	Middle East	2.706	2.172	0.0	0.0	100.0	100.0	46.44	46
BLOOM	Hebrew	Middle East	4.568	1.286	7.34	0.0	96.3	100.0	78.4	79
DictaLM-2.0-7B	Hebrew	Middle East	2.642	2.224	0.0	0.0	83.33	100.0	45.35	45
Gemma-4	Kurdish	Middle East	2.358	2.438	0.0	0.0	100.0	100.0	52.84	53
BLOOM	Kurdish	Middle East	2.446	2.35	0.09	0.0	99.05	100.0	54.81	55
Mistral-Nemo-Kurdish	Kurdish	Middle East	2.486	2.313	0.51	0.0	96.19	100.0	55.7	56
Gemma-4	Azerbaijani	Middle East	2.873	2.619	0.91	0.0	100.0	100.0	54.24	54
BLOOM	Azerbaijani	Middle East	3.296	2.283	0.47	0.0	97.09	100.0	62.23	62
Gemma-4	Uzbek	Middle East	2.839	2.888	0.01	0.0	97.85	100.0	51.09	51
BLOOM	Uzbek	Middle East	2.974	2.757	0.0	0.0	98.92	100.0	53.51	54
Mistral-7B-Uz	Uzbek	Middle East	3.515	2.333	0.0	0.0	100.0	100.0	63.25	64
Gemma-4	Kazakh		3.184	2.452	0.0	0.0	100.0	100.0	54.58	55

tokenizer	language	region	fertility	compression	byte_fallback%	unk%	vcov%	roundtrip%	avg_tok/sent	tot
		Middle East								
BLOOM	Kazakh	Middle East	5.089	1.534	2.79	0.0	86.96	100.0	87.25	882
Gemma-4	Chinese	East Asia	14.132	1.479	0.05	0.0	99.86	100.0	28.89	292
BLOOM	Chinese	East Asia	12.117	1.725	0.79	0.0	98.06	100.0	24.77	250
Gemma-4	Japanese	East Asia	30.432	1.731	0.0	0.0	100.0	100.0	32.51	320
BLOOM	Japanese	East Asia	44.828	1.175	2.32	0.0	87.24	100.0	47.88	482
LLM-jp-3-13B	Japanese	East Asia	25.627	2.055	0.0	0.0	71.3	100.0	27.37	270
Gemma-4	Korean	East Asia	2.415	1.77	0.07	0.0	99.35	100.0	36.83	372
BLOOM	Korean	East Asia	4.887	0.875	22.1	0.0	19.36	100.0	74.52	752
Polyglot-Ko-12B	Korean	East Asia	2.199	1.944	0.13	0.0	93.26	100.0	33.53	330
Gemma-4	Vietnamese	SEA	1.21	3.727	0.18	0.0	100.0	100.0	36.75	372
BLOOM	Vietnamese	SEA	1.128	3.998	0.02	0.0	100.0	100.0	34.26	340
Arcee-VyLinh-3B	Vietnamese	SEA	1.29	3.496	0.89	0.0	100.0	100.0	39.17	390
Gemma-4	Thai	SEA	10.084	2.906	0.0	0.0	100.0	100.0	42.88	430
BLOOM	Thai	SEA	29.119	1.006	11.86	0.0	81.48	100.0	123.81	1230
Typhoon2-7B	Thai	SEA	16.315	1.796	2.35	0.0	98.77	100.0	69.37	702
Gemma-4	Indonesian	SEA	1.575	4.562	0.0	0.0	100.0	100.0	30.83	310
BLOOM	Indonesian	SEA	1.333	5.39	0.0	0.0	99.03	100.0	26.09	260
Nusantara-7B	Indonesian	SEA	2.146	3.347	0.62	0.0	100.0	100.0	42.02	420
Gemma-4	Malay	SEA	1.632	4.437	0.0	0.0	100.0	100.0	32.56	320
BLOOM	Malay	SEA	1.444	5.014	0.0	0.0	100.0	100.0	28.81	290
MaLLaM-5B	Malay	SEA	1.417	5.109	0.04	0.0	84.76	100.0	28.28	280
Gemma-4	Tagalog	SEA	1.84	3.374	0.0	0.0	99.07	100.0	48.66	490
BLOOM	Tagalog	SEA	1.895	3.275	0.02	0.0	98.15	100.0	50.13	500
Gemma-4	Burmese	SEA	6.136	2.499	0.0	0.0	100.0	100.0	64.52	650
BLOOM	Burmese	SEA	25.746	0.596	27.36	0.0	53.59	100.0	270.74	2700
Burmese-GPT-1B	Burmese	SEA	10.719	1.43	0.6	0.0	91.5	100.0	112.72	1140
Gemma-4	Khmer	SEA	12.747	2.067	0.01	0.0	98.38	100.0	73.25	740
BLOOM	Khmer	SEA	29.492	0.893	26.35	0.0	63.78	100.0	169.46	1700
Gemma-4	Swahili	Africa	2.087	3.108	0.0	0.0	100.0	100.0	43.91	440
BLOOM	Swahili	Africa	1.602	4.048	0.05	0.0	100.0	100.0	33.71	340
Swahili-Gemma-7B	Swahili	Africa	2.047	3.169	0.0	0.0	100.0	100.0	43.06	430

tokenizer	language	region	fertility	compression	byte_fallback%	unk%	vcov%	roundtrip%	avg_tok/sent	tot
Gemma-4	Amharic	Africa	3.032	1.658	0.02	0.0	99.08	100.0	51.93	52
BLOOM	Amharic	Africa	7.905	0.636	26.31	0.0	28.0	100.0	135.37	136
Gemma-4	Hausa	Africa	1.864	2.987	0.0	0.0	99.07	100.0	46.33	46
BLOOM	Hausa	Africa	1.924	2.893	1.5	0.0	93.46	100.0	47.84	48
Gemma-4	Yoruba	Africa	2.588	1.944	0.01	0.0	100.0	100.0	64.46	65
BLOOM	Yoruba	Africa	1.759	2.862	0.11	0.0	99.22	100.0	43.8	44
Gemma-4	Igbo	Africa	2.358	2.29	0.0	0.0	100.0	100.0	58.05	58
BLOOM	Igbo	Africa	1.892	2.854	0.15	0.0	100.0	100.0	46.58	47
Kakugo-3B-Igbo	Igbo	Africa	2.688	2.009	4.25	0.0	95.0	100.0	66.19	66
Gemma-4	Zulu	Africa	3.384	2.77	0.0	0.0	100.0	100.0	52.92	53
BLOOM	Zulu	Africa	3.043	3.081	0.01	0.0	99.09	100.0	47.59	48
Gemma-4	Xhosa	Africa	3.302	2.717	0.01	0.0	99.03	100.0	50.6	51
BLOOM	Xhosa	Africa	2.933	3.059	0.03	0.0	99.03	100.0	44.93	45
Gemma-4	Somali	Africa	2.246	2.896	0.0	0.0	100.0	100.0	51.75	52
BLOOM	Somali	Africa	2.402	2.707	0.4	0.0	100.0	100.0	55.36	56
Gemma-4	Wolof	Africa	1.921	2.643	0.02	0.0	99.19	100.0	47.54	48
BLOOM	Wolof	Africa	1.845	2.751	0.01	0.0	100.0	100.0	45.68	46
Wolof-Qwen-1.5B	Wolof	Africa	2.103	2.414	0.53	0.0	100.0	100.0	52.06	52
Gemma-4	Shona	Africa	2.927	2.907	0.0	0.0	100.0	100.0	50.18	50
BLOOM	Shona	Africa	2.848	2.987	0.02	0.0	100.0	100.0	48.83	49
Gemma-4	French	Europe	1.49	4.129	2.04	0.0	98.31	100.0	37.73	38
BLOOM	French	Europe	1.289	4.774	0.0	0.0	100.0	100.0	32.63	33
Lucie-7B	French	Europe	1.427	4.311	0.0	0.0	82.2	100.0	36.13	36
Gemma-4	German	Europe	1.655	4.27	0.01	0.0	100.0	100.0	35.59	36
BLOOM	German	Europe	2.12	3.333	0.05	0.0	100.0	100.0	45.59	46
LeoLM-7B	German	Europe	2.18	3.242	0.01	0.0	81.48	100.0	46.88	47
Gemma-4	Spanish	Europe	1.347	4.47	0.0	0.0	100.0	100.0	34.71	35
BLOOM	Spanish	Europe	1.269	4.745	0.0	0.0	100.0	100.0	32.69	33
Salamandra-7B	Spanish	Europe	1.354	4.447	0.03	0.0	85.58	100.0	34.88	35
Gemma-4	Portuguese	Europe	1.453	4.215	0.0	0.0	100.0	100.0	33.62	34
BLOOM	Portuguese	Europe	1.312	4.669	0.04	0.0	100.0	100.0	30.36	30
Gervasio-8B	Portuguese	Europe	1.709	3.585	0.74	0.0	100.0	100.0	39.53	40
Gemma-4	Italian	Europe	1.535	4.185	0.0	0.0	100.0	100.0	36.87	37
BLOOM	Italian	Europe	1.828	3.514	0.01	0.0	100.0	100.0	43.91	44
LLaMAntino-3-8B	Italian	Europe	1.831	3.508	0.58	0.0	100.0	100.0	43.98	44
Gemma-4	Dutch	Europe	1.63	3.935	0.01	0.0	100.0	100.0	37.05	37
BLOOM	Dutch	Europe	2.04	3.144	0.0	0.0	98.11	100.0	46.37	46

tokenizer	language	region	fertility	compression	byte_fallback%	unk%	vcov%	roundtrip%	avg_tok/sent	tot
Fietje-2	Dutch	Europe	2.325	2.759	0.1	0.0	94.34	100.0	52.83	53
Gemma-4	Polish	Europe	2.096	3.415	1.94	0.0	100.0	100.0	40.37	40
BLOOM	Polish	Europe	2.99	2.394	0.03	0.0	97.17	100.0	57.58	58
Bielik-11B	Polish	Europe	2.924	2.447	3.58	0.0	81.13	100.0	56.32	56
Gemma-4	Russian	Europe	1.884	3.838	0.01	0.0	99.33	100.0	36.58	37
BLOOM	Russian	Europe	3.425	2.111	0.24	0.0	92.67	100.0	66.52	67
Vikhr-Nemo-12B	Russian	Europe	2.13	3.395	0.71	0.0	98.67	100.0	41.36	41
Gemma-4	Ukrainian	Europe	2.273	3.083	0.0	0.0	100.0	100.0	43.09	43
BLOOM	Ukrainian	Europe	3.899	1.798	2.11	0.0	88.44	100.0	73.9	74
MamayLM-12B	Ukrainian	Europe	2.273	3.083	0.0	0.0	100.0	100.0	43.09	43
Gemma-4	Romanian	Europe	1.8	3.479	2.4	0.0	100.0	100.0	42.2	42
BLOOM	Romanian	Europe	2.205	2.84	1.54	0.0	96.43	100.0	51.69	52
LLMic-3B	Romanian	Europe	2.192	2.856	16.17	0.0	77.68	4.15	51.41	52
Gemma-4	Swedish	Europe	1.841	3.543	0.0	0.0	100.0	100.0	36.86	37
BLOOM	Swedish	Europe	2.222	2.935	0.06	0.0	99.05	100.0	44.5	45
Viking-7B	Swedish	Europe	1.474	4.423	0.94	0.0	100.0	100.0	29.53	29
Gemma-4	Czech	Europe	2.157	3.077	1.51	0.0	99.13	100.0	40.87	41
BLOOM	Czech	Europe	2.885	2.3	0.25	0.0	95.65	100.0	54.67	55
CSMPT-7B	Czech	Europe	1.407	4.717	0.03	0.0	96.52	100.0	26.66	26
Gemma-4	Greek	Europe	2.472	2.666	0.0	0.0	100.0	100.0	58.39	59
BLOOM	Greek	Europe	4.341	1.518	1.45	0.0	86.27	100.0	102.52	103
Meltemi-7B	Greek	Europe	1.397	4.72	0.04	0.0	95.42	100.0	32.98	33
Gemma-4	Lat.Am. Spanish	Americas	1.347	4.47	0.0	0.0	100.0	100.0	34.71	35
BLOOM	Lat.Am. Spanish	Americas	1.269	4.745	0.0	0.0	100.0	100.0	32.69	33
LatamGPT-70B	Lat.Am. Spanish	Americas	1.604	3.752	0.65	0.0	100.0	100.0	41.35	41
Gemma-4	Brazilian Portuguese	Americas	1.453	4.215	0.0	0.0	100.0	100.0	33.62	34
BLOOM	Brazilian Portuguese	Americas	1.312	4.669	0.04	0.0	100.0	100.0	30.36	30
Tucano-2b4	Brazilian Portuguese	Americas	1.252	4.893	0.01	0.0	86.84	99.31	28.96	29
Gemma-4	Quechua	Americas	3.147	2.811	0.0	0.0	100.0	100.0	49.58	50
BLOOM	Quechua	Americas	3.158	2.801	0.0	0.0	100.0	100.0	49.74	50
Gemma-4	Haitian Creole	Americas	1.796	2.918	0.0	0.0	100.0	100.0	41.08	41
BLOOM	Haitian Creole	Americas	1.85	2.831	0.05	0.0	99.13	100.0	42.33	42
Gemma-4	Māori	Oceania	1.825	2.647	5.77	0.0	100.0	100.0	54.7	55

tokenizer	language	region	fertility	compression	byte_fallback%	unk%	vcov%	roundtrip%	avg_tok/sent	tot
BLOOM	Māori	Oceania	1.91	2.529	0.02	0.0	94.62	100.0	57.26	579
Gemma-4	Samoan	Oceania	1.821	2.635	0.72	0.0	100.0	100.0	57.27	579
BLOOM	Samoan	Oceania	1.823	2.632	0.02	0.0	97.37	100.0	57.33	580
Gemma-4	Tok Pisin	Oceania	1.644	3.247	0.0	0.0	100.0	100.0	51.44	520
BLOOM	Tok Pisin	Oceania	1.668	3.201	0.0	0.0	100.0	100.0	52.18	520
Airavata-7B	Hindi	Indic	1.823	2.803	3.99	0.0	60.81	100.0	46.2	467
Ambari-7B	Kannada	Indic	2.736	3.133	0.0	0.0	81.69	100.0	43.53	440
MahaMarathi-7B	Marathi	Indic	1.638	4.265	0.0	0.0	84.87	100.0	30.83	311
Qwen-Odia-7B	Odia	Indic	13.654	0.503	25.78	0.0	56.44	75.69	264.72	267
Jais-2-8B	Arabic	Middle East	1.462	4.052	3.71	0.0	86.82	100.0	28.36	287
mGPT-Azerbaijani-1.3B	Azerbaijani	Middle East	3.553	2.118	0.96	0.0	97.09	100.0	67.07	678
KazLLM-8B	Kazakh	Middle East	4.831	1.616	8.27	0.0	91.3	100.0	82.81	838
ChatGLM3-6B	Chinese	East Asia	13.184	1.585	0.02	0.0	10.85	100.0	26.95	272
PrahokBART-62M	Khmer	SEA	14.511	1.816	0.0	8.97	75.14	0.0	83.38	843
Walia-LLM-7B	Amharic	Africa	1.623	3.098	0.03	0.0	88.0	100.0	27.79	287
HausaLlama-8B	Hausa	Africa	2.13	2.614	2.93	0.0	93.46	100.0	52.94	538
YorubaLlama-8B	Yoruba	Africa	2.934	1.715	1.84	0.0	96.09	100.0	73.07	739
Xhosa-ZuluLlama3-8B	Zulu	Africa	3.754	2.497	0.16	0.0	100.0	100.0	58.71	599
Xhosa-ZuluLlama3-8B	Xhosa	Africa	3.567	2.515	0.24	0.0	100.0	100.0	54.65	553
Goldfish-ASM-125M	Assamese	Indic	1.42	4.599	0.0	0.06	93.17	25.89	27.06	273
Goldfish-mri-39M	Māori	Oceania	1.22	3.959	0.12	0.0	92.47	99.7	36.57	370
Goldfish-tpi-125M	Tok Pisin	Oceania	1.311	4.071	0.0	0.04	83.02	98.42	41.02	415
mT5	Hindi	Indic	2.086	2.45	0.0	0.0	54.73	90.32	52.86	534
mT5	Bengali	Indic	2.715	2.436	0.0	0.0	52.47	36.66	52.34	529
mT5	Tamil	Indic	2.541	3.616	0.0	0.0	59.06	92.49	42.12	420
mT5	Telugu	Indic	2.813	2.78	0.0	0.0	59.74	73.02	47.08	470
mT5	Kannada	Indic	3.007	2.852	0.0	0.0	55.63	54.55	47.83	484
mT5	Malayalam	Indic	3.059	3.269	0.0	0.0	62.86	71.15	45.14	450
mT5	Marathi	Indic	2.699	2.588	0.0	0.0	59.21	91.7	50.8	514
mT5	Gujarati	Indic	2.781	2.176	0.0	0.0	56.03	91.8	57.4	580
mT5	Punjabi	Indic	2.743	1.878	0.0	0.0	56.67	43.18	70.21	710
mT5	Odia	Indic	5.33	1.29	0.0	0.0	37.42	64.23	103.33	104
mT5	Assamese	Indic	3.372	1.936	0.0	0.0	55.28	26.38	64.26	650
mT5	Urdu	Indic	1.843	2.549	0.0	0.0	53.33	99.21	50.36	509

tokenizer	language	region	fertility	compression	byte_fallback%	unk%	vcov%	roundtrip%	avg_tok/sent	tot
mT5	Nepali	Indic	2.638	2.572	0.0	0.0	57.79	94.17	48.76	493
mT5	Sinhala	Indic	2.666	2.374	0.0	0.0	51.02	30.43	54.56	552
mT5	Maithili	Indic	2.382	2.208	0.0	0.0	57.24	92.98	57.92	580
mT5	Arabic	Middle East	2.309	2.565	0.0	0.0	53.49	99.9	44.8	453
mT5	Persian	Middle East	1.919	2.742	0.0	0.0	56.59	27.37	44.62	451
mT5	Turkish	Middle East	2.146	3.593	0.19	0.0	55.56	99.31	37.34	371
mT5	Hebrew	Middle East	2.363	2.487	0.0	0.0	62.04	98.32	40.55	410
mT5	Kurdish	Middle East	2.108	2.728	0.0	0.0	58.1	99.41	47.22	471
mT5	Azerbaijani	Middle East	2.383	3.157	0.12	0.0	63.11	99.7	44.99	453
mT5	Uzbek	Middle East	2.556	3.209	0.0	0.0	64.52	99.41	45.99	463
mT5	Kazakh	Middle East	2.336	3.341	0.0	0.0	67.08	98.62	40.05	403
mT5	Chinese	East Asia	14.796	1.413	0.0	0.02	3.35	13.83	30.25	306
mT5	Japanese	East Asia	28.237	1.866	0.0	0.0	4.2	89.23	30.16	305
mT5	Korean	East Asia	2.781	1.537	0.0	0.02	33.58	98.52	42.4	423
mT5	Vietnamese	SEA	2.147	2.101	0.15	0.0	37.71	99.7	65.18	653
mT5	Thai	SEA	7.763	3.775	0.0	0.0	41.36	45.75	33.01	334
mT5	Indonesian	SEA	1.834	3.918	0.0	0.0	59.22	99.11	35.89	363
mT5	Malay	SEA	1.851	3.914	0.0	0.0	58.1	99.7	36.92	373
mT5	Tagalog	SEA	1.838	3.377	0.0	0.0	57.41	99.21	48.62	493
mT5	Burmese	SEA	4.947	3.1	0.0	0.0	50.98	96.25	52.02	523
mT5	Khmer	SEA	8.165	3.227	0.0	0.0	51.89	41.8	46.92	474
mT5	Swahili	Africa	1.972	3.289	0.0	0.0	61.96	99.51	41.5	413
mT5	Amharic	Africa	3.326	1.512	0.0	0.0	29.85	99.31	56.95	570
mT5	Hausa	Africa	1.82	3.06	0.0	0.0	56.07	99.7	45.24	451
mT5	Yoruba	Africa	2.755	1.827	0.01	0.0	53.91	77.47	68.61	694
mT5	Igbo	Africa	2.45	2.204	0.0	0.0	52.5	94.47	60.33	610
mT5	Zulu	Africa	2.99	3.135	0.0	0.0	59.09	99.6	46.76	473
mT5	Xhosa	Africa	2.913	3.08	0.0	0.0	59.22	99.41	44.64	451
mT5	Somali	Africa	2.159	3.012	0.0	0.0	60.4	99.01	49.76	503
mT5	Wolof	Africa	1.957	2.594	0.0	0.0	54.47	99.51	48.44	490
mT5	Shona	Africa	2.619	3.249	0.0	0.0	61.29	99.8	44.9	454

tokenizer	language	region	fertility	compression	byte_fallback%	unk%	vcov%	roundtrip%	avg_tok/sent	tot
mT5	French	Europe	1.849	3.328	0.0	0.0	53.39	96.84	46.8	473
mT5	German	Europe	1.841	3.839	0.0	0.0	58.33	98.91	39.59	400
mT5	Spanish	Europe	1.696	3.549	0.0	0.0	59.62	99.8	43.71	442
mT5	Portuguese	Europe	1.858	3.297	0.0	0.0	53.51	98.42	42.98	434
mT5	Italian	Europe	1.864	3.446	0.0	0.0	56.6	99.6	44.78	453
mT5	Dutch	Europe	1.71	3.75	0.0	0.0	56.6	99.7	38.87	393
mT5	Polish	Europe	2.261	3.166	1.52	0.0	56.6	99.6	43.53	440
mT5	Russian	Europe	2.156	3.353	0.0	0.0	64.67	95.95	41.88	423
mT5	Ukrainian	Europe	2.325	3.015	0.0	0.0	63.27	98.81	44.06	443
mT5	Romanian	Europe	1.938	3.23	1.39	0.0	55.36	99.7	45.45	458
mT5	Swedish	Europe	1.849	3.528	0.0	0.0	60.0	99.01	37.02	374
mT5	Czech	Europe	2.242	2.96	0.43	0.0	54.78	99.01	42.48	423
mT5	Greek	Europe	2.323	2.837	0.0	0.0	62.75	99.01	54.87	553
mT5	Lat.Am. Spanish	Americas	1.696	3.549	0.0	0.0	59.62	99.8	43.71	442
mT5	Brazilian Portuguese	Americas	1.858	3.297	0.0	0.0	53.51	98.42	42.98	434
mT5	Quechua	Americas	3.017	2.932	0.0	0.0	60.0	99.6	47.53	483
mT5	Haitian Creole	Americas	1.772	2.957	0.0	0.0	59.13	97.33	40.52	410
mT5	Māori	Oceania	1.892	2.554	3.89	0.0	62.37	99.8	56.7	573
mT5	Samoan	Oceania	2.029	2.366	0.26	0.0	54.39	99.41	63.8	643
mT5	Tok Pisin	Oceania	1.766	3.024	0.0	0.0	58.49	99.51	55.24	558

7. Raw files

- `data/results.csv` — machine-readable detail (one row per tokenizer × language)
- `data/summary.json` — per-tokenizer aggregates (unweighted + character-weighted)

Regenerate this report: `python experiments/detailed_report.py`